Bioengineered Water Stewardship: Venture-Grade Hackathon Ideation Report

Detecting, Remediating, and Preventing Cooling-System Water Pollution — Track 2

Section I — The 15 Ideas

Idea #1: CoolWater Ontology Engine (CWOE)

Type: Ontology

One-liner: A machine-readable ontology and reasoning layer that formally models every contaminant, pathway, receptor, and regulatory threshold in cooling-water systems so that any downstream tool—sensor, dashboard, or AI agent—can query a single source of truth.

What problem it solves (specific): Today, cooling-water contamination data is siloed across chemical safety data sheets, NPDES permits, vendor-specific biocide formulations, and municipal water quality databases. Engineers cannot programmatically ask "Which contaminant classes from this facility's cooling loop exceed the receiving watershed's aquatic life criteria?" because no structured knowledge graph links chemistry → discharge → ecology → regulation. CWOE closes that gap.

Core approach (how it works): Build an OWL 2 / SKOS ontology with classes for: Contaminant (subclasses: Oxidizing Biocide, Non-Oxidizing Biocide, Corrosion Inhibitor, Scale Inhibitor, Heavy Metal, Microplastic, Thermal Load), Infrastructure Component (cooling tower, blowdown line, basin, make-up water intake), Environmental Receptor (surface water, groundwater, municipal supply), Regulatory Threshold (EPA 304(a) criteria, state-level TMDLs, EU IED BAT-AELs), and Community Impact Indicator (water pressure, turbidity complaints, health incidents). Each contaminant instance is linked to CAS numbers, known ecotoxicological endpoints (LC50, NOEC), typical cooling-water concentration ranges, and approved analytical methods. A SPARQL/reasoning layer (e.g., HermiT OWL reasoner or Apache Jena) enables automated inference: if a facility uses HEDP phosphonate at 15 mg/L in blowdown and the receiving stream has a total phosphorus TMDL of 0.1 mg/L, the reasoner flags a violation risk and links to the relevant permit clause.

Scientific/technical basis: Grounded in IUPAC nomenclature, EPA ECOTOX Knowledgebase toxicity data, ASHRAE Guideline 12 cooling-water chemistry, and CTI (Cooling Technology Institute) standard water quality parameters. Contaminant–pathway–receptor linkage follows the Source-Pathway-Receptor model used in EU Water Framework Directive risk assessment. Reasoning uses OWL 2 DL decidable inference.

MVP / demo (no time framing): A hosted SPARQL endpoint with ≥60 contaminant instances, ≥20 regulatory threshold nodes (US EPA + 3 state TMDLs), and 5 representative facility profiles. Demonstrate: (1) a query returning all contaminants from a hypothetical 5 MW data center cooling loop that exceed aquatic-life criteria in the local receiving water; (2) an automated "risk card" PDF export summarizing findings for a community stakeholder; (3) a REST API returning JSON-LD for downstream integration.

Validation plan + metrics:

1. Ontology coverage completeness: ≥95% of CTI-listed cooling-water additives mapped (target: ≥60 of 63 common chemistries).
2. Reasoning accuracy: ≥98% correct violation/non-violation classification on a 200-case test set of known permit exceedances (precision ≥0.97, recall ≥0.95).
3. Query latency: median SPARQL response <500 ms for single-facility queries on a standard 4-core VM.
4. Interoperability: successful round-trip serialization in OWL/XML, Turtle, and JSON-LD validated by W3C OWL validator.
5. User acceptance: ≥4.0/5 utility rating from 10 pilot reviewers (environmental engineers, regulators).

Feasibility & integration: Requires no physical retrofit—it is a software knowledge layer. Integrates via REST/SPARQL into existing SCADA, BMS, or environmental compliance platforms. Temperature and fouling are not applicable to the software itself, but the ontology models these as operational parameters (e.g., thermal tolerance ranges for bio-based sensors). Failure modes: (1) ontology staleness if regulatory thresholds change and the knowledge base is not updated—mitigated by a quarterly update workflow tied to Federal Register and EU OJ alerts; (2) reasoning errors from incomplete axioms—mitigated by a validation test suite run on every ontology release.

Scalability + cost sketch: Hosting: ~$200/month on a cloud VM with triplestore (e.g., GraphDB free tier or Blazegraph). Ontology maintenance: ~0.25 FTE environmental chemist/year. Marginal cost per additional facility profile: ~2 hours of data entry (~$100 at contractor rates). Scales horizontally; adding a new jurisdiction means adding regulatory threshold nodes (~1 week per new state/country).

Environmental impact model sketch: Baseline: facilities currently spend 40–100 hours/year manually cross-referencing SDS sheets and permits, often missing emerging contaminants. CWOE reduces that to <5 hours/year and catches ≥30% more potential violations before discharge. Inputs: number of facilities, blowdown chemistry profiles, local receiving water criteria. Outputs: estimated avoided permit violations per year, avoided contaminant mass discharged (kg/year), avoided community exposure incidents. Conservative estimate: if 100 facilities adopt and each avoids 1 violation event per year, that prevents ~50,000–500,000 gallons of non-compliant discharge per facility per event.

Market + community viability: Buyer: data center operators (compliance teams), environmental consultancies, municipal water utilities. Pricing: SaaS subscription $500–$2,000/month per facility profile; enterprise license for >50 sites. Revenue path: land 3–5 pilot customers via partnerships with cooling-water treatment vendors (Nalco, ChemTreat, Veolia). Stakeholders: facility operators, state environmental agencies, community water boards. Communication plan: publish ontology under Creative Commons for community trust; offer a free "community query tier" so affected residents can look up local facility risk profiles.

Equity / socioeconomic disparity handling: The free community query tier ensures lower-income communities adjacent to data centers can access the same contaminant–risk information as corporate compliance teams. Risk: if the ontology is only available behind a paywall, it replicates existing information asymmetry. Mitigation: core ontology and community query interface are open-access; revenue comes from enterprise API features (bulk queries, automated reporting, ERP integration).

Regulatory / deployment pathway: Aligns with EPA NPDES electronic reporting (NetDMR), EU IED BAT reporting, and state-level discharge monitoring reports. The ontology can auto-generate data fields for DMR

submissions. No new permits required for the software itself. Deployment: partner with 1–2 state agencies for pilot validation of ontology-driven compliance screening.

Judging scores:

- Technical (35%): 4/5 — Strong knowledge-engineering foundation with concrete reasoning capabilities; slightly limited by being purely informational (no direct measurement).
- Feasibility (25%): 5/5 — Pure software; no physical retrofit; low cost; immediately deployable.
- Environmental Impact (25%): 4/5 — Indirect but significant: prevents violations, enables better monitoring; impact depends on adoption.
- Market/Community (15%): 5/5 — Clear buyer, open-access community tier, strong equity story.
- **Weighted total: (4×0.35)+(5×0.25)+(4×0.25)+(5×0.15) = 1.40+1.25+1.00+0.75 = 4.40/5**

Idea #2: AquaMap — National Cooling-Water Discharge & Risk Atlas

Type: National Mapping

One-liner: A GIS-based national platform that maps every hyperscale and enterprise data center's cooling-water discharge location, overlays receiving water quality, community demographics, and regulatory compliance status into a single interactive risk atlas.

What problem it solves (specific): No centralized, public-facing tool exists that spatially links data center cooling-water discharge points to downstream water quality monitoring stations, TMDL-impaired waterbodies, and Census-tract-level demographic/income data. Regulators, communities, and operators cannot see the cumulative footprint of cooling-water pollution across a region or nation.

Core approach (how it works): Ingest: (1) EPA Facility Registry Service (FRS) and ECHO database for permitted discharge locations; (2) USGS National Water Information System (NWIS) for ambient water quality; (3) EPA 303(d) impaired waters list and TMDL data; (4) Census Bureau ACS 5-year estimates for income, race, and housing data at tract level; (5) EIA-860 and public data center databases (e.g., Baxtel, Data Center Map) for facility locations and estimated cooling loads. Compute a composite "Cooling-Water Stress Index" (CWSI) per facility: f(discharge volume, receiving water assimilative capacity, number of co-located facilities, community vulnerability score based on CDC SVI). Display via a MapLibre/Deck.gl web application with layers for individual facilities, watershed-level aggregation, and community vulnerability heat maps.

Scientific/technical basis: Assimilative capacity modeled using standard mass-balance dilution (C_downstream = C_discharge × Q_discharge / Q_stream + C_ambient × Q_ambient / Q_stream). Community vulnerability uses CDC Social Vulnerability Index (SVI) methodology. Cooling load estimated from publicly reported PUE (Power Usage Effectiveness) and IT load using ASHRAE thermal guidelines (~1.8 L/kWh evaporative loss for wet cooling towers). Data fusion follows EPA EnviroAtlas spatial analysis methods.

MVP / demo (no time framing): A working web map covering ≥3 US states with known hyperscale data center clusters (Virginia, Texas, Oregon). Display ≥50 facility points with discharge metadata, ≥200 downstream NWIS water quality stations, impaired waters overlay, and census-tract SVI layer. Interactive: click a facility to

see estimated blowdown chemistry, receiving water quality, and a "community risk card." Export a CSV of all facilities exceeding a user-defined CWSI threshold.

Validation plan + metrics:

1. Spatial accuracy: ≥95% of facility discharge points within 100 m of verified locations (validated against 30 manually geocoded sites).
2. CWSI correlation: Pearson r ≥ 0.70 between CWSI scores and independently documented community water quality complaints (sourced from state agency complaint databases for ≥2 pilot states).
3. Data completeness: ≥80% of known hyperscale facilities in the 3 pilot states successfully matched to ECHO permits.
4. Latency: map tile load <2 s at national zoom; facility detail query <1 s.
5. User satisfaction: ≥4.2/5 from 15 pilot users (mix of regulators, community advocates, and facility operators).

Feasibility & integration: Pure software/data platform; no physical hardware. Integrates with existing EPA ECHO API, USGS NWIS web services, and Census API—all publicly available. Retrofit not applicable. Failure modes: (1) data lag—EPA ECHO data can be 6–12 months stale; mitigated by supplementing with state-level real-time discharge monitoring data where available; (2) facility misidentification—data center permits may be under parent company names; mitigated by a semi-automated entity resolution pipeline cross-referencing EIA, SEC filings, and Baxtel.

Scalability + cost sketch: Cloud hosting (AWS/GCP): ~$500/month for compute + managed PostGIS. Data ingestion pipeline: ~$300/month for ETL (Airflow + Lambda). Primary cost driver: data curation labor (~0.5 FTE/year for national expansion). Per-state expansion cost: ~$5,000 one-time data curation + $100/month incremental hosting.

Environmental impact model sketch: Baseline: regulators currently review discharge permits facility-by-facility with no cumulative spatial view. AquaMap enables cumulative impact assessment. Inputs: number of facilities per watershed, estimated discharge volumes, ambient water quality. Outputs: number of watersheds exceeding cumulative assimilative capacity, number of communities in high-vulnerability/high-exposure overlap zones. Modeled outcome: identification of ≥10 "hotspot" watersheds nationally where cumulative cooling-water discharge exceeds 5% of low-flow stream assimilative capacity, triggering targeted regulatory review.

Market + community viability: Buyer: (1) state environmental agencies (annual license $10K–$50K); (2) environmental NGOs and community groups (free tier); (3) data center operators wanting pre-development site screening ($5K–$20K/year). Revenue: SaaS tiers. Stakeholders: EPA Office of Water, state DEQs, environmental justice organizations, data center industry associations (e.g., Infrastructure Masons). Comms: launch with a "State of Cooling-Water Discharge" annual report co-branded with an EJ partner.

Equity / socioeconomic disparity handling: The platform's core purpose is to make environmental justice visible. The CWSI explicitly weights community vulnerability (income, race, pre-existing health burden via SVI). Free public access tier ensures communities can self-advocate using the same data regulators see. Risk: data could be misused to stigmatize communities rather than hold polluters accountable. Mitigation: frame all

public outputs around facility responsibility, not community deficiency; partner with EJ organizations for narrative framing.

Regulatory / deployment pathway: Aligns with EPA's EJScreen and EnviroAtlas frameworks. Supports state-level cumulative impact analysis (already mandated in NJ, CA). Could feed into EPA's ECHO Detailed Facility Reports. No permits needed for the software. Deployment: pilot with 1 state DEQ + 1 EJ organization; seek EPA Office of Environmental Justice partnership.

Judging scores:

- Technical (35%): 4/5 — Robust spatial analysis and data fusion; relies on public data (strength and limitation).
- Feasibility (25%): 5/5 — All data sources are public APIs; standard GIS stack; no hardware.
- Environmental Impact (25%): 5/5 — Directly enables cumulative impact assessment and regulatory action; high leverage.
- Market/Community (15%): 5/5 — Strong EJ alignment; clear buyers across public and private sectors.
- **Weighted total: (4×0.35)+(5×0.25)+(5×0.25)+(5×0.15) = 1.40+1.25+1.25+0.75 = 4.65/5**

Idea #3: BioSentinel Inline Probe — Multi-Analyte Cooling-Water Quality Device

Type: Device + Software

One-liner: A ruggedized, pipe-mountable sensor probe combining electrochemical, optical, and immobilized-enzyme elements to continuously measure oxidizing biocides, total heavy metals (Cu/Zn), pH, conductivity, turbidity, and temperature in cooling-water blowdown lines, with edge-compute firmware that streams structured data to a cloud dashboard.

What problem it solves (specific): Cooling-water operators currently rely on grab samples sent to off-site labs (turnaround: 2–7 days) or single-parameter inline sensors (pH, conductivity only). They cannot detect biocide overdosing, heavy metal accumulation from pipe corrosion, or turbidity spikes from flocculant carryover in real time. By the time lab results arrive, non-compliant water has already been discharged.

Core approach (how it works): The probe is a 2-inch NPT threaded stainless-steel housing inserted into the blowdown line via a standard tee fitting. Inside, a multi-sensor cartridge contains: (1) an amperometric free-chlorine/total-oxidant electrode (platinum working electrode, Ag/AgCl reference) for oxidizing biocide detection (target: 0.1–20 mg/L Cl_2-equivalent); (2) an anodic stripping voltammetry (ASV) micro-electrode array (bismuth-film on carbon) for dissolved Cu, Zn, Ni at ppb-level resolution; (3) a nephelometric turbidity sensor (880 nm LED + 90° photodiode, 0–1000 NTU); (4) a 4-electrode conductivity cell (10–100,000 μS/cm); (5) a combination pH electrode (glass membrane, gel-filled reference); (6) an RTD temperature sensor (Pt100, ±0.1°C). An optional enzyme-functionalized membrane layer (organophosphorus hydrolase immobilized on nylon-6,6 membrane) provides a semi-quantitative screen for organophosphonate corrosion inhibitors (HEDP, ATMP) by measuring phosphate released from enzymatic hydrolysis. An ARM Cortex-M4 microcontroller runs on-board signal processing, calibration curves, and anomaly detection (Z-score on 24-h rolling window). Data is transmitted via Modbus RTU to the facility's BMS or via MQTT/4G to a cloud dashboard.

Scientific/technical basis: Amperometric oxidant detection is established (Hach CL17 principle). ASV on bismuth-film electrodes for heavy metals is well-validated in environmental monitoring literature (detection limits ~0.5 ppb for Cu, Zn, Pb). Organophosphorus hydrolase (OPH) has been demonstrated on solid supports for phosphonate/phosphate hydrolysis (Mulchandani et al., various). Nephelometric turbidity follows ISO 7027. All sensing modalities operate in the 15–45°C range typical of cooling-tower blowdown.

MVP / demo (no time framing): A functional bench prototype in a 3D-printed housing demonstrating: (1) real-time chlorine residual tracking in a simulated blowdown stream (tap water + NaOCl at 0.5–10 mg/L); (2) Cu and Zn detection via ASV in spiked samples (10–500 ppb); (3) turbidity response to bentonite additions; (4) data streaming to a Grafana dashboard via MQTT. The enzyme membrane module shown as a proof-of-concept with purified OPH hydrolyzing HEDP in a flow cell.

Validation plan + metrics:

1. Oxidizing biocide accuracy: ±5% of DPD colorimetric reference across 0.1–20 mg/L Cl_2-equivalent (n ≥ 30 paired measurements).
2. Heavy metal detection limit: ≤2 ppb for Cu, Zn; ≤5 ppb for Ni (validated against ICP-MS on ≥20 spiked samples).
3. Turbidity accuracy: ±5% of Hach 2100Q reference across 0–500 NTU.
4. Sensor drift: <5% deviation over 30-day continuous operation in synthetic cooling water (40°C, 2000 μS/cm, pH 8.5).
5. Enzyme membrane activity retention: ≥70% initial activity after 14 days at 35°C in synthetic blowdown.
6. Data uptime: ≥99% data transmission success rate over 30-day field trial.

Feasibility & integration: Mounts via standard NPT fitting on blowdown line—no pipe modification, ~30 min install. Compatible with 15–45°C, pH 6–10, conductivity up to 100,000 μS/cm. Fouling mitigation: automated ultrasonic cleaning cycle every 4 hours; replaceable sensor cartridge every 6 months. Failure modes: (1) biofouling on optical window—mitigated by copper-alloy window surround and ultrasonic burst; (2) ASV electrode passivation in high-hardness water—mitigated by periodic in-situ acid wash cycle; (3) enzyme membrane deactivation—mitigated by modular snap-in replacement ($15 per membrane). Integrates with existing SCADA via Modbus or with cloud platforms via MQTT/REST.

Scalability + cost sketch: BOM for production unit: ~$800 (sensor elements $350, electronics $150, housing $100, assembly/calibration $200). Target sale price: $3,500 per probe + $1,200/year cloud subscription + $180/year for sensor cartridge replacements. Cost per monitored parameter: ~$0.02/measurement (assuming 1-minute intervals). Break-even: ~200 units/year.

Environmental impact model sketch: Baseline: facilities discover blowdown exceedances only via monthly grab samples, resulting in an average 15 days of non-compliant discharge per event. BioSentinel enables detection within minutes, reducing non-compliant discharge by ~95%. Inputs: number of blowdown events/year, typical event duration, contaminant concentrations, discharge volume. Outputs: kg of avoided heavy metal discharge, kg of avoided biocide discharge, gallons of non-compliant water prevented. For a 5 MW facility: estimated 50–200 kg/year avoided Cu+Zn discharge; 500–2,000 kg/year avoided excess biocide discharge.

Market + community viability: Buyer: data center facility managers (opex budget), cooling-water treatment service providers (add to service contracts), third-party environmental compliance firms. Pricing: capital sale + recurring subscription. Why now: ESG reporting pressure, state-level discharge monitoring mandates tightening, Microsoft/Google voluntary water stewardship commitments. Stakeholder engagement: co-develop calibration protocols with ASTM D19 Water committee; offer community-facing read-only dashboard.

Equity / socioeconomic disparity handling: Real-time monitoring data is the single most powerful tool for community accountability. Offer a "Community Transparency Module": a public-facing, anonymized (no proprietary process data) water quality dashboard that any downstream resident can access. Risk: data could cause alarm without context. Mitigation: partner with local water utilities to co-brand and contextualize data (e.g., "blowdown Cu at 50 ppb—below EPA aquatic life criterion of 9 ppb at this hardness" would flag, but data is shown with regulatory context).

Regulatory / deployment pathway: Aligns with EPA Method 334.0 (free chlorine), Method 200.8 (metals by ICP-MS—ASV used as screening, not compliance method, but can trigger compliance sampling). Supports NPDES DMR self-monitoring requirements. Device itself needs no permit; installation may require a minor mechanical permit (plumbing). Path: UL/CSA certification for electrical safety; partner with a certified NELAP lab for method-comparison validation.

Judging scores:

- Technical (35%): 5/5 — Multi-modal sensing with real science; specific analytes, specific methods, bench-validatable.
- Feasibility (25%): 4/5 — Standard fittings, proven sensing; enzyme module adds complexity but is modular/optional.
- Environmental Impact (25%): 5/5 — Direct, quantifiable pollution prevention at the discharge point.
- Market/Community (15%): 4/5 — Clear buyer and revenue; certification pathway needs investment.
- **Weighted total: (5×0.35)+(4×0.25)+(5×0.25)+(4×0.15) = 1.75+1.00+1.25+0.60 = 4.60/5**

Idea #4: CoolGraph — Contamination Causal Knowledge Graph with LLM Query Interface

Type: Ontology / Hybrid (Ontology + AI)

One-liner: A causal knowledge graph encoding cause-effect relationships between cooling-system operating parameters, chemical treatment regimes, contaminant formation pathways, and downstream ecological/community impacts—queryable via a natural-language LLM interface for non-expert stakeholders.

What problem it solves (specific): Cooling-water contamination is a multi-step causal chain (e.g., increased thermal load → higher biocide dosing → biocide breakthrough in blowdown → aquatic toxicity). No existing tool lets a community member, regulator, or junior engineer ask plain-language questions like "If this facility increases its cooling load by 20%, what happens to downstream copper levels?" and get a scientifically grounded, cited answer.

Core approach (how it works): Build a directed acyclic graph (DAG) with nodes representing: Operating Parameters (IT load, PUE, cycles of concentration, blowdown rate), Chemical Inputs (biocide type/dose,

inhibitor type/dose, pH adjusters), Water Quality Parameters (pH, conductivity, TDS, specific contaminant concentrations), Discharge Outcomes (blowdown volume, temperature delta, contaminant mass loading), and Impact Endpoints (aquatic toxicity thresholds, community water quality indicators). Edges encode causal relationships with quantitative transfer functions where data exists (e.g., "Cu pipe corrosion rate ≈ 0.025 mm/yr at pH 8.5, 40°C → dissolved Cu ≈ 150 ppb per 1000 m of Cu pipe at given flow rate") or qualitative direction (increase/decrease) where quantitative data is sparse. An LLM (fine-tuned or RAG-augmented with the graph) provides a chat interface. Queries are decomposed into graph traversals; the LLM synthesizes the traversal result into a human-readable answer with citations to source data.

Scientific/technical basis: Causal DAG structure follows Pearl's causal inference framework. Corrosion rate models from ASHRAE/CTI literature. Biocide dosing–efficacy–breakthrough curves from published CT (concentration × time) data. Ecotoxicological endpoints from EPA ECOTOX database. LLM grounding via retrieval-augmented generation (RAG) with the graph as the retrieval corpus, reducing hallucination.

MVP / demo (no time framing): A web chat interface connected to a Neo4j graph with ≥100 nodes and ≥200 edges covering a "typical evaporative cooling tower" archetype. Demonstrate 5 scenario queries: (1) "What contaminants increase if I raise cycles of concentration from 5 to 8?"; (2) "Is my blowdown copper level likely to exceed the EPA acute freshwater criterion?"; (3) "What happens if I switch from chlorine to bromine-based biocide?"; (4) "Which downstream communities are most vulnerable to my discharge?"; (5) "What monitoring should I add if I start using HEDP inhibitor?" Each answer cites specific graph nodes/edges and underlying data sources.

Validation plan + metrics:

1. Causal accuracy: ≥90% of predicted directional effects (increase/decrease) validated against published cooling-water chemistry studies (test set of 50 cause-effect pairs).
2. Quantitative prediction accuracy: ±25% of measured values for 10 key parameters when compared to field data from 3 published cooling-tower case studies.
3. LLM answer fidelity: ≥95% of cited facts traceable to graph nodes (evaluated by 2 independent reviewers on 50 random queries).
4. Hallucination rate: <5% of answers contain claims not grounded in the graph (same evaluation).
5. User comprehension: ≥80% of non-expert users (n=20) correctly interpret the answer to a standardized test query.

Feasibility & integration: Pure software; Neo4j cloud instance + LLM API. Integrates via API into facility BMS dashboards or standalone web app. No physical retrofit. Failure modes: (1) LLM hallucination despite RAG—mitigated by confidence scoring and "I don't know" fallback when graph coverage is insufficient; (2) graph incompleteness for novel chemistries—mitigated by a structured submission form for expert contributors. Temperature/fouling: not applicable (software).

Scalability + cost sketch: Neo4j Aura: ~$100/month. LLM API costs: ~$0.01–0.05 per query (GPT-4-class via API). Scaling: graph grows with community contributions; LLM costs scale linearly with usage. Total cost for 10,000 queries/month: ~$600/month. Per-query cost decreases with volume.

Environmental impact model sketch: Baseline: facility engineers rely on vendor recommendations and rules of thumb, often overdosing biocides by 20–40% as a safety margin. CoolGraph enables optimized dosing by

making causal trade-offs explicit. Inputs: current dosing practices, CoolGraph-recommended dosing. Outputs: % reduction in biocide usage, associated reduction in blowdown contaminant load. Estimate: 15–25% biocide reduction across adopting facilities → proportional reduction in biocide mass in blowdown.

Market + community viability: Buyer: data center sustainability teams, cooling-water treatment vendors (as a value-add for their customers), environmental consultancies. Pricing: freemium (5 queries/day free; $200/month professional; $1,000/month enterprise with API). Revenue: SaaS. Why now: growing ESG scrutiny of water use; facilities need defensible, data-backed decisions. Stakeholder comms: publish a quarterly "Cooling-Water Causal Insight" digest summarizing common query patterns and emerging risks.

Equity / socioeconomic disparity handling: The natural-language interface is specifically designed to lower the expertise barrier. A community resident can ask questions in plain English without needing to understand cooling-water chemistry. Free tier ensures access. Risk: oversimplified answers could lead to misinterpretation. Mitigation: answers always include a confidence level and a "Consult an expert" flag for high-stakes queries.

Regulatory / deployment pathway: Supports informed public comment on NPDES permits (community members can use CoolGraph to understand permit conditions). Aligns with EPA's Toxics Release Inventory (TRI) community right-to-know mandate. No regulatory approvals needed for the software itself.

Judging scores:

- Technical (35%): 4/5 — Novel causal graph + LLM integration; strong but dependent on graph completeness.
- Feasibility (25%): 5/5 — All software; low cost; existing infrastructure.
- Environmental Impact (25%): 4/5 — Indirect but enables smarter chemical management; depends on adoption.
- Market/Community (15%): 5/5 — Excellent accessibility story; clear SaaS model.
- **Weighted total: (4×0.35)+(5×0.25)+(4×0.25)+(5×0.15) = 1.40+1.25+1.00+0.75 = 4.40/5**

Idea #5: BlowdownGuard — AI-Powered Blowdown Anomaly Detection & Auto-Reporting Platform

Type: Device + Software

One-liner: An edge-AI software module that ingests existing sensor data from cooling-tower controllers (conductivity, pH, flow, temperature) and uses physics-informed ML models to detect anomalous blowdown chemistry, predict contaminant exceedances before they happen, and auto-generate regulatory discharge reports.

What problem it solves (specific): Most cooling towers already have basic sensors (conductivity, pH, temperature, make-up/blowdown flow meters), but the data is used only for simple setpoint control (e.g., "blow down when conductivity exceeds 3,000 μS/cm"). Nobody uses this data stream to infer contaminant concentrations (metals, biocide residuals) or to predict when blowdown will exceed discharge limits. Regulatory reporting is manual, quarterly, and reactive.

Core approach (how it works): Deploy a software agent (runs on existing site server, Raspberry Pi, or cloud) that: (1) ingests real-time data from existing CT controller via Modbus/BACnet; (2) applies a physics-informed neural network (PINN) trained on the relationship between conductivity, pH, temperature, cycles of concentration, and known contaminant behaviors (e.g., Cu solubility as a function of pH and temperature per Pourbaix diagram; biocide decay kinetics as first-order function of temperature and pH); (3) outputs real-time estimated concentrations of 6 "virtual sensors": dissolved Cu, dissolved Zn, total oxidant residual, total phosphorus (from inhibitors), TOC (from biocide breakdown products), and thermal load (ΔT of discharge vs. ambient); (4) flags anomalies using a combination of physics-based thresholds and statistical process control (SPC) charts; (5) auto-generates NPDES-format Discharge Monitoring Reports (DMRs) with estimated values flagged as modeled (not measured) and triggers grab-sample confirmation when exceedances are predicted.

Scientific/technical basis: PINNs incorporate known physical laws as loss-function constraints: (a) Cu/Zn solubility governed by Pourbaix diagrams parameterized by pH, temperature, and dissolved oxygen; (b) biocide decay follows pseudo-first-order kinetics ($k = f(T, pH)$); (c) mass balance on cycles of concentration links make-up chemistry to blowdown chemistry. Training data: synthetic data generated from these physics models + calibration against periodic lab results (quarterly grab samples). The PINN approach is validated in process engineering literature for soft-sensor applications.

MVP / demo (no time framing): Software running on a Raspberry Pi 4, ingesting simulated Modbus data from a Python-based cooling-tower simulator. Dashboard showing: (1) real-time virtual sensor readings for Cu, oxidant, TP, TOC, turbidity, thermal load; (2) SPC charts with control limits and alarm triggers; (3) a "predicted vs. actual" scatter plot using 20 lab-sample calibration points; (4) an auto-generated DMR PDF for a simulated monthly reporting period.

Validation plan + metrics:

1. Virtual sensor accuracy: RMSE ≤15% vs. lab-measured values for Cu, Zn, oxidant, TP across ≥30 paired observations.
2. Anomaly detection sensitivity: ≥90% detection rate for simulated exceedance events (n=50) with false positive rate <10%.
3. Prediction lead time: median ≥4 hours advance warning before a blowdown exceedance reaches the discharge point.
4. DMR generation accuracy: 100% of auto-generated DMR fields match manual DMR for a 6-month historical data set.
5. Compute requirements: runs on Raspberry Pi 4 (4 GB RAM) with <50% CPU utilization at 1-second polling interval.

Feasibility & integration: Requires zero new sensors—piggybacks on existing CT controller data. Software-only installation: ~2 hours to configure Modbus/BACnet polling. Compatible with major CT controllers (Nalco 3D TRASAR, Solenis OnGuard, Chemtreat CHEM-AQUA). Failure modes: (1) physics model mismatch for unusual water chemistries—mitigated by periodic recalibration with lab samples; (2) communication failure with CT controller—mitigated by local data buffering and alert. Temperature: operates in server-room ambient (18–30°C); no exposure to cooling water.

Scalability + cost sketch: Software license: $500/month per cooling tower. Hardware (if needed): $150 for Raspberry Pi + industrial enclosure. Deployment cost: ~$1,000 per site (2 hours install + configuration). Marginal cost per additional tower at same site: $200/month (software only). Scales to thousands of sites with cloud backend.

Environmental impact model sketch: Baseline: facilities detect exceedances only via quarterly lab results, meaning ~90 days of potential non-compliance per event. BlowdownGuard reduces detection latency to hours. Inputs: historical exceedance frequency, blowdown flow rate, contaminant concentrations. Outputs: gallons of non-compliant discharge avoided, kg of contaminant mass avoided. Estimate for a typical 10 MW facility: avoid 4–8 non-compliant blowdown events/year, each lasting 5–30 days → prevent 500,000–3,000,000 gallons of non-compliant discharge/year.

Market + community viability: Buyer: data center operations/facilities team (opex budget line: water treatment). Why now: increasing state requirements for continuous monitoring (e.g., California's proposed electronic reporting mandates); hyperscalers' water stewardship pledges (Microsoft WPP, Google replenishment targets). Revenue: SaaS subscription. Stakeholders: cooling-water treatment companies (channel partners), state regulators (reduced enforcement burden). Comms: case studies showing avoided violations and cost savings.

Equity / socioeconomic disparity handling: Automated, continuous monitoring prevents the "only measure when convenient" dynamic that disproportionately harms downstream communities. Auto-generated DMRs increase transparency. Risk: modeled data could be less accurate than measured data, potentially masking real exceedances. Mitigation: all modeled values include uncertainty bands; any exceedance triggers mandatory grab-sample confirmation within 24 hours; community dashboard shows both modeled and lab-confirmed values.

Regulatory / deployment pathway: Aligns with EPA's push toward electronic DMR submission (NetDMR). Modeled values are flagged as "estimated" per 40 CFR 122.44(i) allowance for monitoring methods. Path to regulatory acceptance: partner with 2–3 state agencies for pilot validation; pursue EPA Environmental Technology Verification (ETV) program recognition.

Judging scores:

- Technical (35%): 5/5 — Physics-informed ML is a rigorous, novel approach with concrete validation.
- Feasibility (25%): 5/5 — No new hardware required; software-only; integrates with existing infrastructure.
- Environmental Impact (25%): 5/5 — Directly prevents non-compliant discharge at scale; quantifiable.
- Market/Community (15%): 4/5 — Strong buyer fit; regulatory acceptance pathway needs development.
- **Weighted total: (5×0.35)+(5×0.25)+(5×0.25)+(4×0.15) = 1.75+1.25+1.25+0.60 = 4.85/5**

Idea #6: ThermoPlume — Satellite + Ground-Truth Thermal Discharge Mapper

Type: National Mapping

One-liner: A national-scale platform that fuses Landsat/Sentinel-2 thermal infrared imagery with ground-truth temperature loggers at discharge points to map, quantify, and track thermal pollution plumes from data center

cooling-water outfalls into receiving waters.

What problem it solves (specific): Thermal pollution from cooling-water discharge is the most volumetrically significant but least monitored impact of data center cooling. Elevated temperature in receiving waters reduces dissolved oxygen, harms aquatic life, and can trigger algal blooms. No existing national platform tracks thermal plumes spatially or temporally for data center outfalls.

Core approach (how it works): (1) Acquire Landsat 8/9 TIRS (thermal infrared sensor, 100 m resolution, 16-day revisit) and Sentinel-3 SLSTR (1 km, daily) imagery for all US/EU waterbodies receiving data center cooling-water discharge. (2) Apply atmospheric correction (split-window algorithm) to derive water surface temperature (WST) maps. (3) Deploy low-cost ground-truth temperature loggers (HOBO TidbiT, ~$80 each) at ≥3 points per site (upstream, discharge point, 500 m downstream) to calibrate satellite-derived WST. (4) Compute thermal plume extent (area where $\Delta T > 1°C$, 2°C, 5°C above ambient) and thermal load ($MW_thermal = \varrho \times Cp \times Q \times \Delta T$). (5) Display on a national web map with time-series animations showing plume evolution. (6) Cross-reference with state thermal discharge limits (typically $\Delta T < 3°C$ or < 5°F above ambient in receiving water).

Scientific/technical basis: Landsat TIRS WST retrieval is well-established (RMSE ~1°C with atmospheric correction). Ground-truth calibration reduces error to ~0.5°C. Thermal plume modeling uses standard mixing-zone equations (CORMIX). Ecological impact thresholds from EPA 304(a) temperature criteria for aquatic life.

MVP / demo (no time framing): A web dashboard showing thermal plume maps for ≥5 data center discharge sites in Virginia (Loudoun County—highest data center density in the world). Time-series of plume extent over 12 months. Ground-truth data from ≥3 sites with co-located HOBO loggers. Automated flagging of exceedances against Virginia's thermal discharge criteria.

Validation plan + metrics:

1. WST accuracy: RMSE ≤0.5°C vs. ground-truth loggers at ≥5 sites (n ≥ 100 satellite-logger paired observations).
2. Plume extent accuracy: ±15% of area compared to high-resolution drone thermal surveys at 2 validation sites.
3. Temporal coverage: ≥80% of cloud-free Landsat passes captured and processed for pilot sites over 12 months.
4. Exceedance detection: ≥95% of ground-truth-confirmed thermal exceedances detected by satellite analysis.
5. Processing latency: satellite-to-map <48 hours from Landsat scene acquisition.

Feasibility & integration: Satellite data is free (USGS Landsat, ESA Sentinel). Ground-truth loggers are $80 each, battery life 5 years, waterproof to 300 m. No facility cooperation required—loggers deployed in public waterways. Software stack: Google Earth Engine or AWS + GDAL/Rasterio. Failure modes: (1) cloud cover limits temporal resolution—mitigated by fusing Landsat + Sentinel-3 for more frequent coverage; (2) satellite spatial resolution insufficient for small streams—mitigated by ground-truth loggers as primary data for small waterways, satellite for large rivers/lakes.

Scalability + cost sketch: Satellite data: free. Ground-truth loggers: ~$400/site (5 loggers). Cloud compute: ~$300/month for national processing. Total to cover 100 sites: ~$40,000 hardware + $3,600/year compute. Per-site marginal cost: ~$400 one-time + ~$36/year.

Environmental impact model sketch: Baseline: thermal discharge is monitored only at the pipe (if at all), not in the receiving water; plume extent is unknown. ThermoPlume provides the first spatial quantification. Inputs: satellite-derived plume areas, temperature differentials, flow data. Outputs: total thermal load (MW) per site, total receiving-water area exceeding thermal criteria (km^2), seasonal trends. Estimate for Virginia alone: ≥20 data center outfalls contributing measurable thermal plumes; aggregate affected stream area ≥5 km^2.

Market + community viability: Buyer: state environmental agencies (thermal TMDL development), environmental NGOs (litigation support), data center operators (pre-siting risk assessment). Pricing: $5,000/year per site for continuous monitoring; $50,000 for state-wide assessment report. Revenue: consulting + SaaS. Stakeholders: USGS, state DEQs, Chesapeake Bay Program (thermal pollution is a Bay impairment factor). Comms: annual "Thermal Footprint of Data Centers" report.

Equity / socioeconomic disparity handling: Thermal pollution disproportionately harms subsistence and recreational fishers in downstream communities. ThermoPlume makes invisible impacts visible. Free public access to thermal plume maps ensures communities can use data in permit hearings and legal actions. Risk: satellite resolution may miss impacts on small headwater streams near facilities—mitigated by prioritizing ground-truth logger deployment in environmental justice communities.

Regulatory / deployment pathway: Supports state thermal TMDL development (Clean Water Act §303(d)). Aligns with EPA thermal discharge guidelines under CWA §316(a). Data format compatible with EPA STORET/WQX for water quality data exchange. No permits needed for remote sensing; logger deployment may need minor waterway access permits.

Judging scores:

- Technical (35%): 4/5 — Solid remote sensing science; limited by satellite spatial/temporal resolution.
- Feasibility (25%): 5/5 — Free data, cheap hardware, standard processing; highly practical.
- Environmental Impact (25%): 5/5 — Addresses the largest-volume cooling impact (thermal pollution) at national scale.
- Market/Community (15%): 4/5 — Clear buyers; niche market but growing with data center expansion.
- **Weighted total: (4×0.35)+(5×0.25)+(5×0.25)+(4×0.15) = 1.40+1.25+1.25+0.60 = 4.50/5**

Idea #7: MicroPlastiSense — Inline Microplastic Particle Counter for Cooling Water

Type: Device + Software

One-liner: A compact flow-through optical particle analyzer that uses laser diffraction and fluorescence (Nile Red staining) to count and size-classify microplastics (≥10 μm) in cooling-tower blowdown, filling a critical monitoring gap for a contaminant class with no existing inline detection method.

What problem it solves (specific): Cooling towers are documented microplastic accumulators and emitters: airborne microplastics are scrubbed from intake air by the water film, concentrated by evaporative cycles, and discharged in blowdown. No inline sensor exists to quantify microplastic loading in cooling water; current methods require grab samples and lab-based FTIR/Raman microscopy (cost: $300–$500/sample, turnaround: 2–4 weeks).

Core approach (how it works): A bypass flow cell (50 mL/min diverted from blowdown line) passes through: (1) an automated Nile Red fluorescent dye injection stage (Nile Red at 1 μg/mL, contact time 5 min in a coiled reactor tube—Nile Red preferentially stains hydrophobic plastic particles); (2) a laser diffraction particle sizer (635 nm diode laser, photodiode array at 0.1–45° scattering angles) for particle size distribution (10–500 μm); (3) a fluorescence detection stage (excitation 470 nm LED, emission filter 580 nm longpass) to discriminate Nile Red-stained plastics from non-plastic particles (minerals, biological matter). The ratio of fluorescent particles to total particles, combined with size distribution, gives microplastic count and size classification. An embedded processor computes: total particle count ≥10 μm, fluorescent (plastic-suspected) particle count, estimated mass concentration (using assumed density 1.05 g/cm^3 for mixed plastics), and size histogram.

Scientific/technical basis: Nile Red staining for microplastic identification is peer-reviewed and widely used in environmental research (Maes et al. 2017, Environmental Science & Technology). Laser diffraction particle sizing is standard (ISO 13320). The combination allows discrimination of microplastics from inorganic particles in complex matrices. Cooling-tower microplastic concentrations have been measured at 5–50 particles/L in recent studies, well within the detection range of this approach.

MVP / demo (no time framing): A benchtop flow cell demonstrating: (1) detection of polyethylene microspheres (50 μm, 100 μm) spiked into synthetic cooling water at 10–100 particles/L; (2) discrimination of PE microspheres from kaolin clay particles via Nile Red fluorescence; (3) real-time particle count and size histogram on a display; (4) data logging to CSV for downstream analysis.

Validation plan + metrics:

1. Particle count accuracy: ±15% vs. manual microscopy count for PE microspheres at 10–500 particles/L (n ≥ 20 samples).
2. Plastic/non-plastic discrimination: ≥85% true positive rate, ≤10% false positive rate for Nile Red fluorescence discrimination (validated with mixed PE + kaolin suspensions, n ≥ 30).
3. Size accuracy: ±10% of certified microsphere diameters (20, 50, 100, 200 μm standards).
4. Detection limit: ≤5 particles/L for ≥50 μm particles (1 L sample volume, 20 min measurement).
5. Dye consumption: <10 mL/day of Nile Red stock solution at continuous operation.

Feasibility & integration: Bypass installation on blowdown line via ¼" tubing and a peristaltic pump. No modification to main process. Operating range: 10–50°C, pH 6–10. Nile Red is non-toxic at μg/mL concentrations; bypass stream returns to blowdown (no added chemical burden). Fouling: inline 5 μm pre-filter (replaceable monthly) removes large debris; flow cell cleaned via automated DI water flush every 12 hours. Failure modes: (1) Nile Red staining of non-plastic hydrophobic particles (e.g., oils)—mitigated by a hydrophobic organic extraction step or by flagging oily samples; (2) biofilm on optical windows—mitigated by UV-C LED exposure during cleaning cycle.

Scalability + cost sketch: BOM: ~$600 (laser diode $50, photodiode array $100, LED/filter $80, peristaltic pump $120, flow cell $50, microcontroller $30, housing $50, Nile Red reagent $20/month). Target sale price: $2,500 + $400/year consumables. Per-measurement cost: ~$0.05. Scales to industrial production; components are commodity.

Environmental impact model sketch: Baseline: zero inline microplastic monitoring exists for cooling towers; total microplastic discharge is unknown. MicroPlastiSense provides the first real-time data. Inputs: measured particle counts × blowdown flow rate. Outputs: estimated microplastic mass discharge (g/day, kg/year) per facility. Enables: (a) optimization of blowdown filtration to reduce microplastic discharge; (b) data for future microplastic discharge regulations. Estimate: a typical cooling tower may discharge 1–10 kg/year of microplastics; monitoring enables >50% reduction through targeted filtration.

Market + community viability: Buyer: data center operators (ESG reporting—microplastics are an emerging reporting metric), cooling-tower equipment manufacturers (value-add), environmental research labs. Pricing: capital sale + consumables. Why now: EU is developing microplastic discharge standards (proposed under Urban Wastewater Treatment Directive revision); US states beginning to study microplastics in industrial discharge. First mover advantage in a market with no existing inline solution. Stakeholders: plastics research community, water utilities, NGOs focused on microplastic pollution.

Equity / socioeconomic disparity handling: Microplastic pollution affects all downstream water users but is invisible without specialized testing. Low-income communities relying on surface water for recreation or subsistence fishing are disproportionately affected. MicroPlastiSense data, shared publicly, arms these communities with evidence. Risk: data anxiety without context (what does "50 particles/L" mean for health?). Mitigation: provide contextual health guidance based on current WHO/EPA assessments alongside raw counts.

Regulatory / deployment pathway: No current US federal microplastic discharge standard, but California SB 1422 mandates microplastic monitoring in drinking water; EU UWWTD revision proposes monitoring in treated effluent. Device positions operators ahead of regulation. No permits needed for the device; data feeds into voluntary ESG reporting frameworks (CDP Water Security, GRI 303).

Judging scores:

- Technical (35%): 4/5 — Novel application of established methods; Nile Red discrimination has limitations.
- Feasibility (25%): 4/5 — Low cost, bypass installation; Nile Red chemistry adds operational complexity.
- Environmental Impact (25%): 4/5 — Addresses an unmonitored contaminant class; impact depends on future regulation.
- Market/Community (15%): 4/5 — First-mover in emerging market; demand depends on regulatory timing.
- **Weighted total: (4×0.35)+(4×0.25)+(4×0.25)+(4×0.15) = 1.40+1.00+1.00+0.60 = 4.00/5**

Idea #8: RegiStream — Automated Cooling-Water Regulatory Compliance & Reporting SaaS

Type: National Mapping / Software

One-liner: A SaaS platform that continuously maps a facility's cooling-water discharge parameters against the applicable federal, state, and local regulatory requirements (NPDES permits, TMDLs, state water quality standards), auto-generates compliance reports, and flags upcoming regulatory changes before they take effect.

What problem it solves (specific): Data center operators must comply with a patchwork of 50+ state-level water quality standards, facility-specific NPDES permit limits, and local ordinances. Compliance teams manually track permit conditions in spreadsheets, often discovering new requirements only after violations. No automated system cross-references real-time (or periodic) discharge data against the full regulatory stack.

Core approach (how it works): (1) Ingest facility discharge data (from BMS, inline sensors, or manual lab uploads). (2) Maintain a structured regulatory database: all 50 US state water quality standards (numeric criteria for >150 parameters), federal EPA 304(a) criteria, facility-specific NPDES permit limits (parsed from permit PDFs via NLP + human QA), and relevant TMDLs. (3) Run continuous compliance checking: compare each discharge parameter against all applicable limits (federal, state, local, permit-specific). (4) Generate: automated DMR pre-fills, compliance status dashboards (green/yellow/red), exceedance alerts (email/SMS/webhook), and regulatory change alerts by monitoring Federal Register and state regulatory agency RSS feeds. (5) Produce annual "Regulatory Fitness Score" per facility.

Scientific/technical basis: Regulatory mapping uses structured data extraction from EPA ECHO, state agency databases, and NLP-based permit parsing (fine-tuned transformer model trained on 500+ NPDES permits for data extraction accuracy). Compliance checking is deterministic rule-based comparison. Regulatory change monitoring uses keyword-filtered Federal Register API and state agency web scrapers.

MVP / demo (no time framing): Platform loaded with: (1) full EPA 304(a) criteria; (2) 3 state water quality standards (VA, TX, OR); (3) 5 parsed NPDES permits from real data center facilities (public records). Demonstrate: upload of a simulated monthly discharge dataset → automated compliance check against all applicable limits → auto-generated DMR PDF → 2 flagged exceedances with corrective action suggestions → regulatory change alert for a proposed VA standard revision.

Validation plan + metrics:

1. Regulatory database coverage: ≥98% of numeric water quality criteria from the 3 pilot states correctly captured (validated against manual review of state code).
2. Permit parsing accuracy: ≥90% of numeric permit limits correctly extracted from PDF permits (validated on 20 test permits).
3. Compliance check accuracy: 100% agreement with manual compliance determination on a 100-case test set.
4. Alert latency: regulatory change alerts issued within 48 hours of Federal Register publication.
5. DMR accuracy: 100% of auto-generated DMR fields match expected values for 10 test datasets.

Feasibility & integration: Pure SaaS; integrates with existing BMS/LIMS via file upload (CSV/Excel) or API. No physical hardware. Failure modes: (1) regulatory database staleness—mitigated by automated monitoring + quarterly human audit; (2) permit parsing errors—mitigated by human-in-the-loop QA for each new permit; (3) false compliance status if input data is wrong—mitigated by data quality checks (range validation, outlier flagging).

Scalability + cost sketch: Cloud hosting: ~$800/month. Regulatory database maintenance: ~0.5 FTE/year ($50K). Per-state expansion: ~$5,000 one-time (data curation) + $500/month incremental. Pricing: $1,000–$5,000/month per facility (depends on permit complexity). Break-even at ~20 facility subscriptions.

Environmental impact model sketch: Baseline: average facility has 2–4 permit exceedance events/year, often discovered weeks to months later. RegiStream enables same-day detection. Inputs: historical violation rates, discharge volumes, contaminant concentrations. Outputs: number of violations prevented, volume of non-compliant discharge avoided, reduction in penalty costs. Estimate: prevent 50–75% of permit violations through early detection → avoid ~$50K–$500K/year in penalties per facility.

Market + community viability: Buyer: data center compliance/legal teams, environmental consultancies managing multiple facilities, cooling-water treatment vendors. Revenue: SaaS subscription. Why now: EPA increasing electronic reporting mandates; states adding new parameters (PFAS, microplastics); ESG investors demanding compliance transparency. Stakeholders: facility operators, state regulators (reduced enforcement burden), environmental law firms.

Equity / socioeconomic disparity handling: Regulatory compliance directly protects downstream communities. The platform's community module provides: a public-facing permit compliance status page for each facility (showing compliance %, recent exceedances, corrective actions). Risk: public-facing data could cause unwarranted alarm. Mitigation: contextualize all data with clear explanations of what limits mean and what exceedances imply for water safety.

Regulatory / deployment pathway: Directly supports NPDES compliance (CWA §402), state water quality certification (§401), and DMR reporting requirements. Aligns with EPA's Next Generation Compliance initiative. No permits needed for the software. Deployment: partner with 2–3 environmental law firms and 1 state DEQ for pilot.

Judging scores:

- Technical (35%): 4/5 — Solid regulatory informatics; NLP-based permit parsing is technically strong.
- Feasibility (25%): 5/5 — Pure SaaS; no hardware; direct integration with existing workflows.
- Environmental Impact (25%): 4/5 — Prevents violations; impact is indirect (enablement, not direct remediation).
- Market/Community (15%): 5/5 — Clear buyer, urgent pain point, strong revenue model.
- **Weighted total: (4×0.35)+(5×0.25)+(4×0.25)+(5×0.15) = 1.40+1.25+1.00+0.75 = 4.40/5**

Idea #9: EnzymeLoop — Immobilized Phosphonate-Degrading Enzyme Cartridge for Blowdown Treatment

Type: Bio-Remediation / Device Hybrid

One-liner: A modular, cartridge-based bioremediation unit containing immobilized alkaline phosphatase and organophosphorus hydrolase on ceramic monolith supports that enzymatically degrades phosphonate corrosion inhibitors (HEDP, ATMP, PBTC) in cooling-tower blowdown before discharge, converting them to orthophosphate and organic fragments amenable to conventional treatment.

What problem it solves (specific): Phosphonate-based corrosion/scale inhibitors (HEDP, ATMP, PBTC) are the most widely used scale inhibitors in cooling water. They are poorly biodegradable (half-lives of months in the environment), contribute to eutrophication of receiving waters, and are not removed by standard blowdown treatment (typically just pH adjustment and settling). Total phosphorus from phosphonate discharge often exceeds TMDL allocations, especially in phosphorus-impaired watersheds (e.g., Chesapeake Bay watershed).

Core approach (how it works): A cylindrical cartridge (15 cm diameter × 50 cm length) containing a cordierite ceramic honeycomb monolith (400 cpsi, 2 mm channel width) with enzyme immobilized on the channel walls via glutaraldehyde cross-linking to aminopropyl-triethoxysilane (APTES)-functionalized ceramic surface. Two enzyme layers: (1) organophosphorus hydrolase (OPH, EC 3.1.8.1) cleaves the C-P bond in HEDP/ATMP, releasing orthophosphate; (2) alkaline phosphatase (ALP, EC 3.1.3.1) further hydrolyzes any residual organophosphate esters. Blowdown water flows through the monolith channels at 1–5 L/min (contact time ~30 seconds). Orthophosphate in the effluent is then removed by a downstream conventional step (chemical precipitation with alum or iron salts, already present in many blowdown treatment systems).

Scientific/technical basis: OPH has well-characterized activity against organophosphonate substrates (Vmax, Km values published for HEDP). Immobilization on ceramic supports via APTES-glutaraldehyde is established in industrial enzyme technology (activity retention typically 40–70%). Ceramic monoliths tolerate 15–60°C, pH 6–10, and the conductivity range of cooling water. ALP is commercially available and stable at slightly alkaline pH (optimal pH 8–10, matching cooling-water conditions). The degradation pathway converts recalcitrant C-P bond compounds into treatable inorganic phosphorus.

MVP / demo (no time framing): A bench-scale cartridge (5 cm × 20 cm) with OPH + ALP immobilized on a cordierite monolith section. Demonstrate: (1) ≥60% HEDP degradation in a single pass at 0.5 L/min flow rate, 35°C, pH 8.5; (2) orthophosphate release confirmed by molybdenum blue colorimetric assay; (3) enzyme activity stability over 72 hours of continuous operation; (4) downstream orthophosphate removal to <0.5 mg/L by alum addition.

Validation plan + metrics:

1. HEDP degradation efficiency: ≥60% per pass at 10 mg/L HEDP inlet, 35°C, pH 8.5, 1 L/min flow (measured by ICP-OES for total P and ion chromatography for orthophosphate).
2. Enzyme operational half-life: ≥30 days at 35°C under continuous flow (activity measured weekly via standard pNPP assay for ALP, paraoxon assay for OPH).
3. Monolith pressure drop: <5 kPa at 5 L/min (measured with differential pressure gauge).
4. Total phosphorus in treated effluent: <0.5 mg/L after enzymatic + alum polishing step.
5. Fouling rate: <10% flow reduction over 14 days without cleaning (measured by flow rate at constant pressure).

Feasibility & integration: Cartridge inserts into blowdown line via standard flanged connection, between the cooling tower basin and the discharge/treatment system. Sized for 1–50 L/min blowdown flow (multiple cartridges in parallel for larger towers). Temperature range: 15–50°C (enzyme activity decreases above 50°C but cooling-water blowdown rarely exceeds 45°C). Maintenance: cartridge replacement every 3–6 months (depending on enzyme half-life); simple swap-out. Failure modes: (1) enzyme deactivation by oxidizing biocides—mitigated by installing cartridge downstream of dechlorination step (standard in blowdown

treatment); (2) monolith clogging from suspended solids—mitigated by a 50 μm strainer upstream; (3) incomplete degradation—downstream alum step provides a safety net.

Scalability + cost sketch: Cartridge BOM: ~$200 (ceramic monolith $50, enzyme preparation $100, immobilization reagents $30, housing $20). Target sale price: $600/cartridge, replaced every 4–6 months = $1,200–$1,800/year. Treatment cost: ~$0.05–$0.10/m^3 of blowdown. For comparison, conventional phosphorus chemical precipitation alone costs ~$0.02–$0.05/m^3 but cannot address the recalcitrant organophosphonate fraction.

Environmental impact model sketch: Baseline: a facility using 15 mg/L HEDP with blowdown at 50 m^3/day discharges 0.75 kg/day total phosphorus. With EnzymeLoop achieving 60% degradation + alum polishing, discharge TP drops to <0.1 kg/day. Inputs: blowdown flow, inlet HEDP concentration. Outputs: kg/day TP removed, annual TP reduction, equivalent reduction in eutrophication potential (kg PO_4-eq using CML characterization factors). Estimate per facility: 200–400 kg/year TP reduction.

Market + community viability: Buyer: data center facilities with phosphorus-impaired receiving waters (Chesapeake Bay states, Great Lakes region, EU Danube basin). Pricing: cartridge sales + optional monitoring subscription. Why now: Chesapeake Bay TMDL enforcement tightening; EPA considering lowering recommended phosphorus criteria. Stakeholders: cooling-water treatment vendors (white-label the cartridge), water utilities, environmental groups focused on eutrophication.

Equity / socioeconomic disparity handling: Eutrophication from phosphorus discharge degrades recreational water quality, fisheries, and property values—disproportionately affecting waterfront communities with fewer resources to adapt. EnzymeLoop directly reduces this burden. Risk: cartridge waste disposal—mitigated by designing for ceramic monolith recycling (enzyme can be stripped and re-immobilized).

Regulatory / deployment pathway: Supports NPDES permit compliance for total phosphorus limits. Aligns with Chesapeake Bay TMDL implementation plans. Cartridge is a treatment technology, not a chemical addition—simplifies permitting (no new chemical introduction to the discharge stream). Path: pilot at 2–3 facilities in VA/MD with TP permit limits → seek Virginia DEQ validation → pursue NSF/ANSI 61 certification if treated water contacts potable systems.

Judging scores:

- Technical (35%): 4/5 — Strong enzyme biochemistry; immobilization is proven; specific target analyte.
- Feasibility (25%): 4/5 — Cartridge-based, modular; needs upstream dechlorination; 4-6 month replacement.
- Environmental Impact (25%): 5/5 — Direct, measurable phosphorus removal from a recalcitrant contaminant.
- Market/Community (15%): 4/5 — Clear niche (P-impaired watersheds); market size depends on regulatory pressure.
- **Weighted total: (4×0.35)+(4×0.25)+(5×0.25)+(4×0.15) = 1.40+1.00+1.25+0.60 = 4.25/5**

Idea #10: CoolTwin — Digital Twin Simulator for Cooling-Water Chemistry & Discharge

Type: Device + Software (Software-heavy, pairs with any sensor)

One-liner: A physics-based digital twin of a cooling-tower water cycle that simulates the full chemistry of make-up, recirculation, chemical treatment, evaporation, drift, and blowdown—enabling operators to predict discharge quality under any operating scenario before it happens.

What problem it solves (specific): Operators cannot predict how changes in operating parameters (increased IT load, changed chemical treatment, seasonal temperature shifts, make-up water quality changes) will affect blowdown discharge chemistry. Decisions are made reactively based on historical rules of thumb, leading to over-treatment (waste) or under-treatment (discharge violations).

Core approach (how it works): A deterministic simulation model of the cooling-tower water cycle: (1) make-up water composition (from utility water quality reports or inline sensor input); (2) evaporation model (Merkel equation for evaporative cooling, function of wet-bulb temperature, air flow, heat load); (3) concentration model (cycles of concentration = make-up flow / blowdown flow, with drift losses); (4) chemical equilibrium model (PHREEQC-based speciation for Ca, Mg, SiO_2, pH, alkalinity at tower conditions—predicts scaling potential, metal solubility, biocide speciation); (5) chemical treatment model (biocide decay kinetics, inhibitor adsorption/depletion, pH adjustment); (6) blowdown composition calculator (output: full discharge chemistry for every parameter). The user inputs: IT load (MW), ambient weather (API from NOAA), make-up water quality (manual or API), chemical treatment regime. The twin outputs: predicted blowdown composition at 1-hour resolution; compliance status against permit limits; recommended adjustments (e.g., "reduce biocide dose by 15% because lower ambient temperature reduces microbial growth rate").

Scientific/technical basis: Merkel equation for cooling-tower thermal performance is ASHRAE standard. PHREEQC is USGS's open-source geochemical speciation code, validated for decades. Biocide kinetics based on published CT (concentration × time) models. Corrosion rate estimation using Langelier/Ryznar indices. The integration of these established models into a single digital twin for cooling-water discharge prediction is novel.

MVP / demo (no time framing): A web application where the user configures a virtual cooling tower (capacity, fill type, make-up water source) and a chemical treatment program. Run a 30-day simulation under historical weather data (NOAA API). Display: time-series of all blowdown parameters, compliance traffic lights, scaling/corrosion risk indices, and 3 "what-if" scenarios (e.g., "what if IT load increases 30%?", "what if we switch from chlorine to chloramine?", "what if make-up quality degrades due to drought?").

Validation plan + metrics:

1. Blowdown TDS prediction: ±10% vs. measured TDS at ≥3 real cooling-tower sites (n ≥ 30 measurements).
2. pH prediction: ±0.3 pH units vs. measured.
3. Scaling index prediction: correct scaling/non-scaling classification in ≥90% of cases vs. observed scale deposits.
4. Biocide residual prediction: ±20% vs. measured free chlorine/total oxidant.
5. Scenario response accuracy: "what-if" predictions for 5 parameter changes validated against published case studies.

Feasibility & integration: Pure software; runs in browser or on-premise. Can ingest data from existing sensors via API or manual input. No physical hardware needed. Failure modes: (1) model inaccuracy for unusual water chemistries (e.g., high silica)—mitigated by PHREEQC's extensive mineral database; (2) weather forecast errors—mitigated by showing confidence intervals based on forecast uncertainty; (3) user input errors—mitigated by range checks and default values.

Scalability + cost sketch: Development: PHREEQC (free, open-source) + custom Python/JS wrapper. Hosting: ~$300/month for compute (PHREEQC runs are lightweight). Pricing: $300–$1,500/month per facility. Marginal cost per additional facility: negligible (software).

Environmental impact model sketch: Baseline: operators use fixed chemical dosing rates, leading to ~20–40% over-treatment on average. CoolTwin enables dynamic dosing optimization. Inputs: current vs. optimized chemical doses. Outputs: kg/year reduction in biocide discharge, kg/year reduction in inhibitor discharge, m^3/year water saved through optimized blowdown. Estimate: 15–30% reduction in chemical discharge, 5–10% reduction in blowdown volume.

Market + community viability: Buyer: data center facility managers, cooling-water treatment vendors (as a premium service offering), engineering consultancies. Revenue: SaaS. Why now: digital twin adoption accelerating across industry; ESG reporting demands quantitative water stewardship data. Stakeholders: ASHRAE TC 3.6, CTI, facility management platforms (Honeywell, Schneider Electric).

Equity / socioeconomic disparity handling: Predictive optimization reduces chemical discharge that harms downstream communities. The "what-if" scenario feature lets community advocates explore: "What happens to blowdown if this facility doubles in size?" and present findings in permit hearings. Community access via a read-only scenario mode (no proprietary facility data exposed). Risk: model confidence could be over-stated. Mitigation: always display prediction uncertainty ranges.

Regulatory / deployment pathway: Supports permit applications by providing modeled discharge projections for proposed facility expansions. Aligns with EPA's Environmental Modeling Community of Practice. Can generate "reasonable worst-case" discharge estimates for NPDES permit limit derivation.

Judging scores:

- Technical (35%): 5/5 — Integrates established, validated models (PHREEQC, Merkel) in a novel application.
- Feasibility (25%): 5/5 — Pure software; open-source foundation; no hardware.
- Environmental Impact (25%): 4/5 — Enables optimization; impact depends on operator action.
- Market/Community (15%): 4/5 — Strong buyer fit; needs differentiation from generic process simulation.
- **Weighted total: (5×0.35)+(5×0.25)+(4×0.25)+(4×0.15) = 1.75+1.25+1.00+0.60 = 4.60/5**

Idea #11: BioFilm Trap — Engineered Biofilm Reactor for Heavy Metal Capture from Blowdown

Type: Bio-Remediation

One-liner: A packed-bed bioreactor seeded with metal-hyperaccumulating biofilm (Pseudomonas putida engineered to overexpress metallothionein and extracellular polysaccharide production) that passively captures dissolved Cu, Zn, and Ni from cooling-tower blowdown to below permit limits, with periodic biofilm harvesting for metal recovery.

What problem it solves (specific): Copper, zinc, and nickel leach from cooling-system metallurgy (Cu condenser tubes, galvanized steel, Ni alloys) at concentrations of 50–500 ppb—often exceeding EPA aquatic life criteria (Cu: 3–18 ppb depending on hardness; Zn: 120 ppb; Ni: 52 ppb). Conventional treatment (chemical precipitation) is effective only above ~1 mg/L and generates large volumes of chemical sludge. A biological approach can achieve lower residual concentrations with less chemical input.

Core approach (how it works): A cylindrical packed-bed reactor (30 cm diameter × 100 cm height) filled with ceramic Raschig rings (25 mm) provides ~150 m^2/m^3 surface area for biofilm attachment. The reactor is inoculated with a consortium of: (1) Pseudomonas putida KT2440 (biosafety level 1, well-characterized, robust biofilm former) overexpressing smtA (metallothionein from Synechococcus) for intracellular Cu/Zn sequestration; (2) wild-type Shewanella oneidensis MR-1 (produces extracellular polymeric substances that chelate metals). Blowdown flows upward through the reactor at 5–20 L/min (hydraulic retention time ~15 minutes). Biofilm sorbs dissolved metals via: (a) extracellular polysaccharide binding (rapid, minutes); (b) intracellular metallothionein accumulation (hours); (c) biomineralization on biofilm surface (days). The reactor effluent meets target metal concentrations. Every 60–90 days, the biofilm is harvested (reactor drained, biofilm sloughed by air scouring), and the metal-rich biomass is sent for recovery (acid leaching) or disposal.

Scientific/technical basis: P. putida KT2440 is a model organism for environmental biotechnology with extensive genetic tools. Metallothionein overexpression in P. putida has been demonstrated to increase Cu biosorption 3–5× (Valls et al. 2000). Shewanella's EPS production and metal-binding capacity are well-documented. Biofilm reactors for heavy metal removal achieve residual concentrations of 1–10 ppb in published studies at similar hydraulic loading rates. Temperature tolerance: P. putida grows at 15–42°C; Shewanella at 4–37°C (adequate for blowdown at 25–40°C).

MVP / demo (no time framing): A bench-scale column (5 cm × 30 cm) with ceramic packing inoculated with wild-type P. putida KT2440 + S. oneidensis (non-engineered for MVP, to avoid BSL-2 requirements). Demonstrate: (1) ≥70% Cu removal from 200 ppb Cu synthetic blowdown at 0.5 L/min, 35°C, pH 8.0; (2) effluent Cu < 20 ppb; (3) biofilm establishment confirmed by crystal violet staining and confocal microscopy; (4) metal mass balance (inlet load - effluent load = biofilm-associated metal, confirmed by acid digest of harvested biofilm + ICP-MS).

Validation plan + metrics:

1. Cu removal efficiency: ≥70% from 200 ppb inlet → <60 ppb outlet at 35°C, pH 8.0, 15 min HRT (n ≥ 10 steady-state samples).
2. Zn removal efficiency: ≥60% from 300 ppb inlet.
3. Biofilm establishment time: steady-state performance within 14 days of inoculation.
4. Operational stability: <15% variation in removal efficiency over 60-day continuous operation.
5. Metal recovery: ≥80% of biofilm-associated metals recoverable by 0.1 M HCl wash.
6. Containment: zero colony-forming units of inoculated organisms detected in effluent (confirmed by selective plating on rifampicin-supplemented agar).

Feasibility & integration: Installs as a side-stream treatment unit on the blowdown line (similar footprint to a conventional sand filter). Operates at cooling-water temperature and pH without adjustment. Nutrient requirement: minimal—cooling water contains sufficient organic carbon (TOC 5–50 mg/L) and nutrients for biofilm maintenance; supplemental carbon (citrate, 5 mg/L) added if TOC < 5 mg/L. Failure modes: (1) biofilm washout from flow surges—mitigated by hydraulic surge protection (equalization tank) and using high-surface-area packing; (2) biocide carryover killing the biofilm—mitigated by dechlorination upstream (sodium thiosulfate injector, standard practice); (3) containment breach (engineered organisms in effluent)—mitigated by downstream UV disinfection and use of auxotrophic strains (cannot survive without supplemented nutrients absent in the environment).

Scalability + cost sketch: Reactor BOM: ~$1,500 (vessel $500, packing $300, plumbing $200, inoculum $200, instrumentation $300). Operating cost: ~$50/month (nutrients, monitoring). Target price: $5,000/unit + $600/year service contract. Treatment cost: ~$0.02/m³. For comparison, ion exchange treatment: ~$0.10–$0.50/m³.

Environmental impact model sketch: Baseline: facility discharges 200 ppb Cu in 50 m³/day blowdown = 10 g/day = 3.65 kg/year Cu to receiving water. BioFilm Trap reduces to <60 ppb = <3 g/day = <1.1 kg/year. Net reduction: ~2.5 kg/year Cu per facility. Inputs: blowdown flow, inlet metal concentrations. Outputs: kg/year metals removed, metal recovery value, comparison to chemical precipitation sludge volume.

Market + community viability: Buyer: data centers with metal-limited NPDES permits (common in Cu-impaired watersheds). Pricing: capital sale + service contract. Why now: EPA updating Cu aquatic life criteria based on Biotic Ligand Model (BLM)—many facilities will face tighter limits. Stakeholders: cooling-tower equipment manufacturers (offer as accessory), metal recyclers (biofilm-harvested metal), state agencies.

Equity / socioeconomic disparity handling: Heavy metal discharge disproportionately affects communities relying on local fisheries and surface water. BioFilm Trap directly reduces this exposure. Metal recovery creates minor economic value that could partially fund community water monitoring programs. Risk: public concern about "engineered bacteria"—mitigated by using BSL-1 organisms, closed reactor design, and downstream UV disinfection; community education on containment.

Regulatory / deployment pathway: Technology is a treatment unit, regulated under state wastewater treatment permit (not a new chemical). Engineered-organism version requires EPA TSCA review for commercial use of genetically modified microorganisms; wild-type consortium avoids this. Path: pilot with wild-type consortium → validate performance → pursue EPA ETV → develop engineered consortium for enhanced performance under TSCA notification.

Judging scores:

- Technical (35%): 4/5 — Strong microbiology; proven mechanisms; containment is well-addressed.
- Feasibility (25%): 3/5 — Biofilm maintenance adds operational complexity; dechlorination required; 60-90 day harvest cycles.
- Environmental Impact (25%): 5/5 — Direct metal removal from blowdown; quantifiable kg/year reduction.
- Market/Community (15%): 3/5 — Niche market (metal-limited permits); public acceptance of bio-reactor needs work.

- **Weighted total: (4×0.35)+(3×0.25)+(5×0.25)+(3×0.15) = 1.40+0.75+1.25+0.45 = 3.85/5**

Idea #12: WaterEquity Dashboard — Community Impact & Environmental Justice Scoring Platform

Type: National Mapping / Software

One-liner: A public-facing web platform that computes and displays a "Cooling-Water Equity Score" for every community within 10 km of a data center cooling-water discharge point, integrating water quality data, socioeconomic vulnerability indicators, and health outcome data to quantify disproportionate burden.

What problem it solves (specific): Environmental justice analyses for data center water impacts are ad hoc, inconsistent, and rarely performed proactively. Communities adjacent to data centers often experience water quality degradation but lack data to demonstrate disproportionate impact in regulatory proceedings. No standardized scoring system exists for cooling-water-specific EJ assessment.

Core approach (how it works): (1) Identify all data center cooling-water discharge points (from EPA FRS, state permits, public databases). (2) For each discharge point, define an "impact zone" (10 km downstream in the receiving watershed, modeled using NHDPlus flowlines). (3) For each Census tract within the impact zone, compute a "Cooling-Water Equity Score" (CWES) combining: (a) Water Quality Burden sub-score (proximity to discharge, number of 303(d) impairments, frequency of reported water quality complaints—from state complaint databases); (b) Socioeconomic Vulnerability sub-score (CDC SVI: income, education, race/ethnicity, linguistic isolation, housing burden); (c) Health Susceptibility sub-score (CDC PLACES data for asthma, kidney disease, cardiovascular disease rates at tract level); (d) Cooling Infrastructure Intensity sub-score (number of facilities within 20 km, estimated aggregate cooling-water discharge volume). Each sub-score is normalized 0–100; CWES = weighted average. Display on a national map with tract-level granularity.

Scientific/technical basis: Methodology mirrors EPA's EJScreen but is specific to cooling-water impacts. NHDPlus is the USGS authoritative hydrologic dataset. SVI is CDC's validated vulnerability index. PLACES provides modeled health estimates at tract level using BRFSS data. Weighting of sub-scores determined by expert elicitation (Delphi method with EJ researchers, water quality scientists, and community advocates).

MVP / demo (no time framing): A working web map for Virginia and Texas showing: (1) ≥50 data center discharge points; (2) CWES scores for all Census tracts within 10 km of any discharge; (3) interactive drill-down showing sub-score components; (4) "Community Risk Report" PDF export for any tract; (5) comparison tool: "how does my community's score compare to state average?"

Validation plan + metrics:

1. Spatial accuracy: 100% of impact zones correctly delineated along NHDPlus flowlines (validated against manual delineation for 10 test sites).
2. CWES reproducibility: intra-rater reliability ≥0.95 (same inputs always produce same score).
3. CWES face validity: ≥80% agreement between CWES rankings and expert panel rankings for 20 test communities (Spearman $\rho \geq 0.80$).

4. Data currency: all sub-score data sources updated within 12 months of latest release (ACS, SVI, PLACES).
5. Community utility: ≥4.0/5 utility rating from 15 community advocates in pilot testing.

Feasibility & integration: Pure web application using public data APIs (Census, CDC, EPA). No hardware. Failure modes: (1) data gaps in rural areas—mitigated by flagging low-confidence scores where PLACES or complaint data is sparse; (2) score misinterpretation—mitigated by extensive contextual help text and community training resources.

Scalability + cost sketch: Development: standard web stack (React + PostGIS + Python). Hosting: ~$400/month. Data refresh: automated quarterly. Per-state expansion: ~$3,000 one-time data curation. Total national expansion: ~$150K one-time + $5K/month ongoing.

Environmental impact model sketch: Baseline: EJ analysis for cooling-water impacts is performed only when communities or NGOs invest significant resources. WaterEquity Dashboard democratizes this analysis. Inputs: number of communities assessed, CWES scores, identified disparities. Outputs: number of high-burden communities identified, number of regulatory actions triggered by CWES data. Impact: identification of ≥50 "high-burden" communities nationally that warrant priority regulatory attention.

Market + community viability: Buyer: (1) environmental NGOs and EJ organizations (foundation-funded, free access); (2) state environmental agencies (license for integration into permit review workflows, $10K–$30K/year); (3) data center developers (pre-siting risk screening, $5K–$15K/year). Revenue: mixed (grant-funded free tier + government/corporate paid tiers). Why now: Biden-era EJ Executive Order 14096; state-level EJ screening mandates (CA, NJ, WA); growing ESG scrutiny of data center water use.

Equity / socioeconomic disparity handling: This *is* an equity tool. It is designed to quantify and make visible the disproportionate burden of cooling-water pollution on vulnerable communities. Risk: the tool itself could be used by developers to avoid high-score areas, potentially concentrating data centers in "lower-score" areas that are already economically disadvantaged for other reasons. Mitigation: the scoring methodology accounts for pre-existing vulnerability, so any new facility near a vulnerable community would raise the score; publish a "responsible siting" guide alongside the tool.

Regulatory / deployment pathway: Aligns with EPA EJScreen, Council on Environmental Quality's Climate and Economic Justice Screening Tool (CEJST), and state-level EJ screening tools. Can be referenced in NPDES permit public comment periods. Supports Title VI civil rights complaints. No permits needed.

Judging scores:

- Technical (35%): 3/5 — Sound methodology but primarily data integration, not novel science or technology.
- Feasibility (25%): 5/5 — Public data, standard web technology, immediately deployable.
- Environmental Impact (25%): 4/5 — Enables action but doesn't directly reduce pollution; high social impact.
- Market/Community (15%): 5/5 — Directly serves communities; strong alignment with current EJ policy momentum.
- **Weighted total: (3×0.35)+(5×0.25)+(4×0.25)+(5×0.15) = 1.05+1.25+1.00+0.75 = 4.05/5**

Idea #13: QuatSense — Quaternary Ammonium Compound Biosensor Patch

Type: Device + Software

One-liner: A disposable, postage-stamp-sized biosensor patch containing a whole-cell E. coli reporter strain engineered with a QAC-responsive promoter driving GFP expression, coupled to a smartphone-readable fluorescence detector, for rapid semi-quantitative screening of quaternary ammonium compound (QAC) biocides in cooling-water blowdown and receiving waters.

What problem it solves (specific): Quaternary ammonium compounds (quats—e.g., alkyl dimethyl benzyl ammonium chloride, ADBAC) are among the most common non-oxidizing biocides in cooling water. They are highly toxic to aquatic organisms (LC50 for fish: 0.5–5 mg/L) and can persist in discharge. No rapid, low-cost field test exists for QAC detection; current methods require HPLC (cost: $100+/sample, lab only).

Core approach (how it works): An E. coli BL21 strain carrying a plasmid with the qacR regulon from Staphylococcus aureus: qacR is a TetR-family repressor that de-represses in the presence of QACs. The qacR-responsive promoter drives GFP. When QACs are present, qacR de-represses, GFP is expressed, and fluorescence is detectable within 2–4 hours. The cells are lyophilized (freeze-dried) onto a nitrocellulose pad in a sealed foil pouch. To use: add 200 μL of sample, incubate at 30–37°C (body heat or portable incubator), and read fluorescence with a smartphone-attached UV LED + filter clip accessory. Software converts fluorescence intensity to estimated QAC concentration (semi-quantitative: <0.1, 0.1–1, 1–10, >10 mg/L).

Scientific/technical basis: The qacR/qacA system is one of the best-characterized QAC resistance regulons (Brown & Skurray, 1999). qacR binds a broad range of QACs with Kd values of 0.1–10 μM—well within the relevant concentration range for cooling-water blowdown (typical 1–50 mg/L active QAC). Whole-cell biosensors with transcription-factor-driven fluorescent reporters are well-established in synthetic biology (van der Meer, 2004). Lyophilization of biosensor strains is commercially practiced (e.g., Aptasensor strips).

MVP / demo (no time framing): (1) E. coli BL21 strain carrying pUC19-qacR-Pqac-GFP plasmid. (2) Demonstrate dose-response: GFP fluorescence measured by plate reader at 0, 0.1, 1, 5, 10 mg/L ADBAC in synthetic cooling water (pH 8, 2000 μS/cm, 35°C). (3) Show smartphone readout with UV LED clip producing semi-quantitative classification. (4) Show lyophilized patches retain >50% activity after 30-day room-temperature storage.

Validation plan + metrics:

1. Detection limit: ≤0.1 mg/L ADBAC (confirmed by ≥3 σ above background fluorescence, n ≥ 10).
2. Dynamic range: 0.1–50 mg/L QAC, with $R^2 \geq 0.90$ for log-linear calibration curve.
3. Specificity: <20% cross-reactivity with non-QAC biocides (chlorine, isothiazolone, glutaraldehyde) at 10 mg/L.
4. Response time: detectable fluorescence within 3 hours at 37°C.
5. Shelf life: ≥50% activity retention after 90-day storage at 25°C in sealed foil pouch.
6. Smartphone accuracy: ±1 classification level (e.g., "1–10 mg/L" vs. "0.1–1 mg/L") compared to plate reader, in ≥80% of tests.

Feasibility & integration: No installation—it's a grab-sample, single-use test. Can be used by operators, regulators, or community members. No power needed for incubation (body heat or $20 USB hand warmer). Smartphone readout requires a $15 UV LED + filter clip (3D-printable). Failure modes: (1) false positives from cooling-water matrix effects (surfactants, high ionic strength)—mitigated by a negative control patch included in each test kit; (2) temperature sensitivity—mitigated by including a temperature-indicating strip on the patch; (3) biosafety concerns with live E. coli—mitigated by using a kill-switch strain (essential gene under conditional control) and providing a bleach inactivation step post-read.

Scalability + cost sketch: BOM per patch: ~$0.50 (nitrocellulose $0.05, freeze-dried cells $0.20, pouch $0.10, reagents $0.15). Smartphone clip: ~$15 (one-time per user). Target sale price: $5/patch (pack of 20: $80). Scale: standard fermentation + lyophilization; contract manufacturing feasible at 10,000+ patches/month.

Environmental impact model sketch: Baseline: QAC monitoring is virtually non-existent for cooling-water blowdown; discharge concentrations are unknown. QuatSense enables: (a) first-ever field surveillance of QAC in cooling-water discharges; (b) identification of over-dosing events; (c) data for future QAC discharge standards. Inputs: number of measurements per site × QAC concentrations detected. Outputs: distribution of QAC discharge concentrations nationally; number of sites exceeding proposed aquatic life thresholds. Estimate: with adoption at 100 facilities × 12 measurements/year = 1,200 data points/year, creating the first national QAC discharge dataset.

Market + community viability: Buyer: (1) community water monitoring groups (citizen science—funded by EJ grants, ~$80/pack); (2) cooling-water treatment vendors (QC tool, ~$200/pack of 50); (3) state regulators (field screening during inspections). Revenue: consumable sales. Why now: EPA is reviewing QAC environmental toxicity following COVID-era surge in antimicrobial use; data gap is well-recognized. Stakeholders: EPA Office of Pesticide Programs (QACs are registered pesticides), academic environmental chemistry groups, citizen science networks.

Equity / socioeconomic disparity handling: At $5/test, QuatSense is accessible to community groups. It empowers residents to independently verify whether cooling-water discharge contains biocides, without needing $100+ lab tests. Risk: community alarm from detecting any QAC. Mitigation: each kit includes an interpretation guide with ecological thresholds and contextual guidance; partner with Extension Services for community training.

Regulatory / deployment pathway: QACs are EPA-registered pesticides under FIFRA; discharge into waterways is regulated under CWA. QuatSense data can support state enforcement actions. The biosensor device itself is exempt from EPA registration (it's a detection tool, not a pesticide). Biosafety: E. coli BL21 is BSL-1; lyophilized format minimizes exposure; kill-switch for commercial version addresses Release concerns under EPA's TSCA biotechnology framework.

Judging scores:

- Technical (35%): 5/5 — Novel biosensor with specific, well-characterized mechanism; concrete molecular biology.
- Feasibility (25%): 3/5 — Requires lyophilization, kill-switch development, smartphone clip; field validation needed.
- Environmental Impact (25%): 4/5 — Fills a critical data gap for an unmonitored contaminant class.

- Market/Community (15%): 4/5 — Low cost, citizen-science-friendly; market development needed.
- **Weighted total: (5×0.35)+(3×0.25)+(4×0.25)+(4×0.15) = 1.75+0.75+1.00+0.60 = 4.10/5**

Idea #14: SpectraFlow — Portable UV-Vis-Fluorescence Spectrometer for Multi-Contaminant Cooling-Water Screening

Type: Device + Software

One-liner: A handheld, USB-C-powered spectrometer (UV-Vis absorbance 200–800 nm + fluorescence at 3 excitation wavelengths) coupled with a machine-learning chemometric model that provides simultaneous semi-quantitative screening of 8+ cooling-water contaminant classes from a single 5 mL sample in under 60 seconds.

What problem it solves (specific): Cooling-water operators and field inspectors need rapid multi-analyte screening but current options are either single-parameter (test strips, single-channel colorimeters) or require expensive lab instruments (ICP, HPLC, IC). No portable device provides simultaneous screening across biocides, metals, inhibitors, and organics.

Core approach (how it works): A compact spectrometer module (based on CMOS linear array detector, Ocean Insight STS-VIS form factor) with: (1) deuterium + tungsten light source for UV-Vis absorbance (200–800 nm, 1 nm resolution); (2) three LED excitation sources for fluorescence (280 nm for protein/aromatic organics, 365 nm for NADH/humic-like, 470 nm for chlorophyll/algae). A 5 mL sample is pipetted into a 10 mm quartz flow-through cuvette. In 60 seconds, the device acquires full UV-Vis absorbance and 3-channel fluorescence emission spectra. An on-device ML model (partial least squares regression + random forest ensemble, trained on a library of >500 synthetic cooling-water mixtures at varying concentrations) outputs estimated concentrations for: (a) oxidizing biocides (free chlorine/bromine via UV absorbance at 290–330 nm); (b) non-oxidizing biocides (isothiazolone/glutaraldehyde via UV absorbance at 220–280 nm); (c) dissolved Cu (absorbance at 600–900 nm for Cu^{2+} aquo complex); (d) dissolved Zn (indirect, via zinc-specific chelator included in a reagent pad); (e) total organic carbon proxy (UV absorbance at 254 nm, "UV254"); (f) phosphonate inhibitors (phosphomolybdate reaction with reagent pad, absorbance at 880 nm); (g) turbidity (side-scatter at 860 nm); (h) algal growth (chlorophyll-a fluorescence at 680 nm emission).

Scientific/technical basis: UV-Vis spectral deconvolution for multi-analyte water quality is well-established (s::can spectro::lyser principle). UV254 as a TOC proxy is US EPA-endorsed. Phosphomolybdate method for phosphorus is standard (SM 4500-P). PLS regression for spectral deconvolution of complex mixtures has extensive literature support. Miniature spectrometers (Ocean Insight, Hamamatsu) achieve 1–2 nm resolution in a handheld form factor.

MVP / demo (no time framing): A benchtop unit using an Ocean Insight STS-VIS spectrometer module + 3 LEDs + cuvette holder + Raspberry Pi running the ML model. Demonstrate: (1) calibration curves for chlorine (0.1–10 mg/L), glutaraldehyde (0.5–50 mg/L), Cu (0.05–5 mg/L), UV254 (0.01–1 cm^{-1}), turbidity (0–100 NTU), PO_4 (0.1–10 mg/L), and chlorophyll-a (0.5–50 μg/L); (2) prediction of concentrations in 10 "unknown" synthetic cooling-water mixtures; (3) comparison to reference methods for each analyte.

Validation plan + metrics:

1. Chlorine accuracy: ±10% vs. DPD reference across 0.1–10 mg/L (n ≥ 20).

2. Cu accuracy: ±20% vs. ICP-MS across 0.05–5 mg/L (n ≥ 20).
3. UV254/TOC proxy: $R^2 \geq 0.85$ vs. combustion TOC analyzer (n ≥ 30).
4. PO_4 accuracy: ±10% vs. ion chromatography across 0.1–10 mg/L (n ≥ 20).
5. Multi-analyte prediction (blind test): ≥75% of 8 analytes within ±25% of true concentration in 10 mixed samples.
6. Measurement time: <90 seconds from sample introduction to results display.

Feasibility & integration: Handheld, battery-powered (USB-C, 4-hour battery life). No installation. Used for grab-sample analysis in the field. Requires $10 reagent packs (Zn chelator pad + phosphomolybdate pad) per 50 measurements. Failure modes: (1) matrix interference from unknown cooling-water additives—mitigated by including a "model confidence" score that flags low-confidence predictions; (2) cuvette contamination—mitigated by DI rinse between samples; (3) spectral overlap—mitigated by PLS deconvolution + inclusion of common interferents in training library.

Scalability + cost sketch: BOM: ~$500 (spectrometer module $200, LEDs + optics $100, cuvette $30, Raspberry Pi + battery $100, housing $70). Target sale price: $2,000 + $10/reagent pack. Per-measurement cost: ~$0.20. Production: outsource to PCBA/optics contract manufacturer; MOQ ~100 units.

Environmental impact model sketch: Baseline: field inspectors can currently test only 1–2 parameters per sample with portable equipment. SpectraFlow tests 8 simultaneously, enabling comprehensive screening in the field. Inputs: number of field measurements × number of parameters per measurement. Outputs: improvement in parameter coverage per inspection, number of additional exceedances detected. Estimate: 4× increase in parameters screened per field visit → 2–3× increase in exceedance detection rate.

Market + community viability: Buyer: state environmental inspectors, cooling-water treatment service technicians, data center facility teams, environmental consultancies. Revenue: instrument sales + consumable reagent packs. Why now: miniature spectrometer costs have dropped 70% in 10 years; ML-based chemometrics now run on edge devices. Stakeholders: Ocean Insight (spectrometer OEM), ASTM D19 Water committee (standard method development), EPA field operations.

Equity / socioeconomic disparity handling: At $2,000, SpectraFlow is within reach of community water monitoring groups (many have budgets of $5K–$20K). Enables citizen-science-grade multi-analyte screening previously requiring $50K+ lab equipment. Risk: semi-quantitative results could be misinterpreted as definitive. Mitigation: results display includes "screening only—confirm with certified lab analysis" disclaimer; training materials for community groups.

Regulatory / deployment pathway: Positions as a field-screening tool, not a compliance monitoring instrument. Aligns with EPA's Quick Guide to Drinking Water Sample Collection for screening applications. Data supports "reasonable potential" determinations in NPDES permits. Path to compliance method status: submit method validation package to EPA Office of Water for alternate test procedure approval under 40 CFR 136.

Judging scores:

- Technical (35%): 4/5 — Proven spectroscopic principles; ML chemometrics strong but semi-quantitative.
- Feasibility (25%): 4/5 — Off-the-shelf components; handheld form factor; needs reagent packs.

- Environmental Impact (25%): 4/5 — Enables comprehensive field screening; multiplies monitoring capacity.
- Market/Community (15%): 4/5 — Broad buyer base; affordable; needs method validation for regulatory acceptance.
- **Weighted total: (4×0.35)+(4×0.25)+(4×0.25)+(4×0.15) = 1.40+1.00+1.00+0.60 = 4.00/5**

Idea #15: AlgaeShield — Immobilized Microalgae Panel for Passive Nutrient & Metal Polishing of Blowdown

Type: Bio-Remediation / Hybrid

One-liner: Modular panel units containing immobilized Chlorella vulgaris in alginate beads that passively remove residual phosphorus, nitrogen, and dissolved metals from cooling-tower blowdown through algal uptake, operating as a "living treatment wetland in a box" that requires only sunlight or LED illumination.

What problem it solves (specific): After primary blowdown treatment, residual nutrients (P, N from inhibitors and biocide decomposition) and trace metals remain at concentrations above ecological thresholds but below levels treatable by conventional chemical methods. Constructed wetlands are effective but require large land areas. A compact, modular algal treatment panel provides the same biological polishing in a small footprint.

Core approach (how it works): Flat panel bioreactors (1 m × 0.5 m × 5 cm) filled with 3% sodium alginate beads (3 mm diameter) containing immobilized Chlorella vulgaris at 10^8 cells/mL. Blowdown flows through the panel at 1–2 L/min (HRT ~10 min). Chlorella removes: (a) PO_4 via luxury phosphorus uptake (polyphosphate body storage); (b) NH_4^+/NO_3^- via assimilation; (c) dissolved Cu, Zn, Ni via cell-surface adsorption and intracellular accumulation. Panels are placed in a rack (like solar panels) with optional LED illumination for indoor/low-light deployment. Alginate bead matrix prevents cell washout and allows easy harvesting (drain panel, remove spent beads, reload fresh beads every 30–60 days). Spent beads are composted (nutrients recovered) or anaerobically digested (biogas + nutrient recovery).

Scientific/technical basis: Chlorella vulgaris is the most-studied alga for wastewater treatment; phosphorus uptake rates of 5–15 mg P/g dry weight/day are well-documented. Alginate immobilization of Chlorella is extensively published (Mallick 2002; Ruiz-Marin & Mendoza-Espinosa 2008), with advantages in cell retention and ease of handling. Cu biosorption by Chlorella: 20–60 mg/g dry weight. Operating pH 7–9 and temperature 15–40°C are compatible with cooling-water conditions.

MVP / demo (no time framing): A bench-scale panel (20 cm × 20 cm × 3 cm) with Chlorella-alginate beads. Demonstrate: (1) ≥50% PO_4 removal from 5 mg/L inlet at 0.1 L/min, 30°C, under 100 μmol/m^2/s LED light; (2) ≥40% Cu removal from 100 ppb inlet; (3) bead integrity over 14-day continuous flow; (4) algal viability (Fv/Fm chlorophyll fluorescence ≥ 0.6) throughout operation.

Validation plan + metrics:

1. PO_4 removal: ≥50% from 5 mg/L inlet (effluent <2.5 mg/L) at steady state (n ≥ 10 samples over 14 days).
2. Cu removal: ≥40% from 100 ppb inlet (effluent <60 ppb) at steady state.
3. NH_4 removal: ≥30% from 2 mg/L inlet.

4. Algal viability: Fv/Fm ≥ 0.5 after 30 days continuous operation.
5. Bead integrity: <5% bead breakage over 30 days (assessed by visual inspection + particle count in effluent).
6. Light requirement: demonstrate effective operation under both natural sunlight (outdoor) and LED (indoor, 100–200 μmol/m²/s PAR).

Feasibility & integration: Panels install in a rack downstream of primary blowdown treatment. Footprint: ~2 m² per 5 L/min flow capacity. Indoor operation: LED panels ($50/m², 20 W/m²). Nutrient supplementation: not needed (blowdown has sufficient N, P, and trace elements). Failure modes: (1) algal die-off from biocide carryover—mitigated by dechlorination upstream; (2) temperature extremes—mitigated by insulated panel housing and optional heating element ($30); (3) alginate bead dissolution at high pH (>10)—mitigated by pH monitoring and blowdown pH adjustment (typically already at 8–9).

Scalability + cost sketch: Panel BOM: ~$150 (acrylic panel $40, alginate beads $30, Chlorella inoculant $20, plumbing $30, LED strip $30). Target price: $400/panel + $50/bead reload kit. Treatment cost: ~$0.03/m³. Scales by adding panels in parallel. For a 10 m³/day blowdown: ~4 panels = $1,600 + $200/month bead reloads.

Environmental impact model sketch: Baseline: after primary treatment, blowdown still contains 1–3 mg/L PO_4, 30–100 ppb Cu. AlgaeShield polishes to <1.5 mg/L PO_4, <60 ppb Cu. Inputs: blowdown flow, post-treatment concentrations. Outputs: kg/year additional P, N, metal removed by algal polishing. Estimate per facility (50 m³/day blowdown): 30–50 kg/year additional P removed; 0.5–1 kg/year additional Cu removed. Harvested algal biomass: ~200–500 kg dry weight/year—usable as compost or anaerobic digestion feedstock.

Market + community viability: Buyer: data centers with stringent nutrient limits (Great Lakes, Chesapeake Bay, Gulf Coast hypoxia zones), municipal wastewater utilities co-treating cooling-water blowdown. Revenue: panel sales + bead reload subscriptions. Why now: nutrient TMDLs are the most commonly litigated water quality limits; EPA pushing for more stringent P limits. Stakeholders: algae biotech companies (potential white-label), water utilities, agricultural compost operators (for spent biomass).

Equity / socioeconomic disparity handling: Algae-based treatment is a visible, tangible, "green" technology that builds community trust more than chemical treatment. Spent algal biomass can be donated to community gardens as compost—creating direct local benefit. Risk: "open algae" perception issues—mitigated by enclosed panel design, no release of algal cells to environment (alginate immobilization), and community education.

Regulatory / deployment pathway: Treatment technology; permitted under state wastewater treatment permits. No genetically modified organisms (wild-type Chlorella vulgaris). Spent biomass disposal: compostable per state solid waste regulations. Aligns with EPA's interest in nature-based solutions for water treatment. Path: pilot at 2 data center sites → EPA ETV → commercial launch.

Judging scores:

- Technical (35%): 4/5 — Well-established algal biology; alginate immobilization proven; concrete targets.
- Feasibility (25%): 3/5 — Needs light (LED cost), bead reloads, dechlorination; moderate maintenance.
- Environmental Impact (25%): 4/5 — Direct nutrient/metal removal; secondary benefit of biomass recovery.
- Market/Community (15%): 4/5 — Green tech appeal; growing market driven by nutrient TMDLs.

- **Weighted total: (4×0.35)+(3×0.25)+(4×0.25)+(4×0.15) = 1.40+0.75+1.00+0.60 = 3.75/5**

Section II — Top 5 Ranked Winners

(Ranked by weighted total; tie-break by Feasibility, then Technical)

Rank 1: Idea #5 — BlowdownGuard (Weighted total: 4.85/5)

- **Technical:** Physics-informed neural networks grounded in Pourbaix diagrams, first-order biocide kinetics, and mass balance—a rigorous, scientifically novel approach to "virtual sensing" with concrete validation against lab measurements.
- **Feasibility:** Zero new hardware; piggybacks entirely on existing cooling-tower controller data streams; installs in 2 hours; runs on a $150 Raspberry Pi.
- **Environmental Impact:** Directly prevents non-compliant blowdown discharge by providing hours of advance warning; quantifiable impact (500K–3M gallons of non-compliant discharge avoided/year per 10 MW facility).
- **Market/Community:** Clear buyer (facility operations team), strong ROI narrative (avoided violations + penalties), and auto-generated DMRs reduce compliance burden.
- **Equity:** Continuous monitoring eliminates the "only measure when convenient" dynamic; community dashboard included.

Rank 2: Idea #2 — AquaMap (Weighted total: 4.65/5)

- **Technical:** Robust spatial data fusion of EPA, USGS, Census, and industry databases with a novel Cooling-Water Stress Index.
- **Feasibility:** 100% public data, standard GIS stack, immediately deployable with no hardware.
- **Environmental Impact:** Enables the first national-scale cumulative impact assessment of cooling-water discharge—a paradigm shift for regulatory and community awareness.
- **Market/Community:** Strong multi-sector buyer base (agencies, NGOs, operators); EJ-centered design; free public access tier.
- **Equity:** Core purpose is making environmental justice visible; explicitly designed to serve vulnerable communities.

Rank 3 (tie-break: Feasibility 5, Technical 5): Idea #10 — CoolTwin (Weighted total: 4.60/5)

- **Technical:** Integrates ASHRAE Merkel equation, USGS PHREEQC geochemical speciation, and biocide kinetics into a single digital twin—novel synthesis of validated models.
- **Feasibility:** Pure software; open-source PHREEQC foundation; no hardware; browser-based; extremely low cost.

- **Environmental Impact:** Enables predictive chemical dosing optimization, reducing chemical discharge by an estimated 15–30%.
- **Market/Community:** Digital twins are a well-understood concept for data center operators; clear SaaS model.
- **Equity:** "What-if" scenarios empower community advocates to model expansion impacts.

Rank 4 (tie-break: Feasibility 4, Technical 5): Idea #3 — BioSentinel Inline Probe (Weighted total: 4.60/5)

- **Technical:** Multi-modal sensing (amperometric, ASV, nephelometric, enzymatic) with specific analytes, specific methods, and concrete detection limits—the strongest hardware science in the portfolio.
- **Feasibility:** Standard NPT installation, 30-min install, 6-month cartridge replacement; enterprise-grade.
- **Environmental Impact:** Real-time, multi-analyte monitoring at the discharge point directly prevents non-compliant discharge.
- **Market/Community:** Clear capital-sale + subscription model; community transparency module included.
- **Equity:** Community-facing dashboard with regulatory context provides downstream accountability.

Rank 5: Idea #6 — ThermoPlume (Weighted total: 4.50/5)

- **Technical:** Established remote sensing science (Landsat TIRS) calibrated with low-cost ground-truth loggers; addresses the most under-monitored cooling impact (thermal pollution).
- **Feasibility:** Free satellite data, $80 loggers, standard cloud compute; extraordinarily cost-effective.
- **Environmental Impact:** Provides the first spatial quantification of thermal plumes from data center discharges nationally; enables thermal TMDL development.
- **Market/Community:** Clear buyers (state DEQs, NGOs); annual "Thermal Footprint" report drives awareness.
- **Equity:** Thermal pollution harms subsistence fishers disproportionately; ThermoPlume makes invisible impacts visible.

Section III — Portfolio Coverage Check

- **Ontology:** Covered by Idea #1 (CWOE — full OWL 2 ontology) and Idea #4 (CoolGraph — causal knowledge graph with LLM interface). Both provide structured knowledge foundations with distinct architectures (formal ontology vs. causal DAG).
- **National Mapping:** Covered by Idea #2 (AquaMap — discharge risk atlas), Idea #6 (ThermoPlume — thermal plume mapper), Idea #8 (RegiStream — regulatory compliance mapping), and Idea #12 (WaterEquity Dashboard — EJ scoring platform). Multiple mapping perspectives: chemical, thermal, regulatory, and social.
- **Device + Software:** Covered by Idea #3 (BioSentinel — multi-analyte inline probe), Idea #5 (BlowdownGuard — AI-powered anomaly detection), Idea #7 (MicroPlastiSense — microplastic

counter), Idea #13 (QuatSense — biosensor patch), and Idea #14 (SpectraFlow — portable spectrometer). Range from inline to handheld to disposable.

- **Monitoring deficit:** Addressed by all device ideas (#3, #5, #7, #13, #14) and mapping platforms (#2, #6) — collectively covering real-time inline, field screening, remote sensing, and regulatory reporting gaps.
- **Chemical complexity:** Addressed by ideas targeting specific contaminant classes: oxidizing biocides (#3, #5, #14), QACs (#13), phosphonates (#9, #14), heavy metals (#3, #5, #11, #14, #15), microplastics (#7), and thermal load (#6). The ontology (#1) and causal graph (#4) model chemical complexity systemically.
- **Infrastructure lock-in:** All ideas explicitly address retrofit constraints: software/mapping tools require zero physical changes; device ideas use standard fittings (NPT), bypass lines, or portable/disposable formats; bio-remediation ideas (#9, #11, #15) install as modular side-stream units with no modification to the main cooling loop.
- **Regulatory lag:** Directly addressed by Idea #8 (RegiStream — automated compliance and regulatory change tracking) and supported by Idea #1 (CWOE — regulatory threshold nodes), Idea #2 (AquaMap — permit compliance overlay), and Idea #12 (WaterEquity — EJ regulatory support). The portfolio collectively closes the gap between current regulation and emerging contaminants.
- **Equity / socioeconomic disparity:** Every idea includes an explicit equity strategy. Idea #12 (WaterEquity Dashboard) is dedicated entirely to EJ. Free community access tiers are included in #1, #2, #4, #6, and #12. Low-cost community tools (#13 at $5/test, #14 at $2,000) democratize monitoring. Community-facing data dashboards (#3, #5, #6) enable accountability.